

YOLO Model Architecture, Accuracy Benchmarks & Extensibility Guide

Academic Reference Document for EviGuard AI Proctoring System

1. Comparative Accuracy & Speed Benchmarks (YOLO vs Other Models)

In real-time edge applications such as online exam proctoring, a vision model must achieve high mean Average Precision (mAP) while maintaining a high frame rate (>25 FPS) on commodity hardware without excessive GPU reliance. Below is a comprehensive comparative evaluation against alternative computer vision architectures:

Model Architecture	Type / Paradigm	COCO mAP (50-95)	Inference (CPU / GPU)	AI Proctoring Viability
YOLO26 Nano (Ultralytics)	Single-Stage (NMS-Free)	38.5%	30–45 FPS (CPU) 120+ FPS (GPU)	Ideal — Zero latency, ultra-lightweight (3.2M params).
YOLO26 / v11 Small	Single-Stage (Anchor-Free)	46.8%	20–30 FPS (CPU) 95+ FPS (GPU)	High Accuracy — Robust small object detection.
Faster R-CNN (ResNet-50)	Two-Stage (RPN + RoI)	40.2%	4–8 FPS (CPU) 18–25 FPS (GPU)	Unsuitable — Excessive lag; frames drop in live stream.
SSD MobileNet v2	Single-Stage (Anchor-Based)	22.1% – 31.2%	30–40 FPS (CPU) 70+ FPS (GPU)	Suboptimal — Fails to detect partially occluded phones.
RT-DETR (Baidu / Ultralytics)	Vision Transformer (DETR)	53.1% – 54.8%	3–6 FPS (CPU) 35–50 FPS (GPU)	GPU Only — High precision, but requires heavy VRAM.

2. Technical Advantages of YOLO26 in AI Proctoring

- 1. Single-Stage Regression (Zero Latency):** Unlike two-stage detectors that generate regional proposals before classification, YOLO26 formulates object detection as a unified spatial regression problem with native NMS-free end-to-end inference (<10ms).
- 2. Anchor-Free Head Architecture:** YOLO26 uses an anchor-free task-aligned assigner that predicts the center and dimensions directly. This significantly enhances detection of rotated smartphones, notebooks, and shifting candidate postures.
- 3. CPU and Edge Deployment:** The lightweight backbone (C2f feature fusion blocks) enables smooth 30 FPS inference on standard laptops without requiring a dedicated CUDA GPU.
- 4. Multi-Task Ecosystem:** Ultralytics YOLO26 provides a unified API supporting Object Detection, Pose Estimation, Hand Keypoint Tracking, and Instance Segmentation under a consistent interface.

3. Implementing YOLO26 in EviGuard

Model Lineage & Architecture: The EviGuard proctoring engine natively integrates **YOLO26** via the Ultralytics framework. The codebase is engineered with a decoupled **Factory Pattern Architecture** (DetectorFactory) that guarantees 100% plug-and-play compatibility across all YOLO versions.

How YOLO26 is Integrated in EviGuard:

Step 1: Checkpoints (yolo26n.pt and yolo26n-pose.pt) reside in the root workspace.

Step 2: config.yaml configures the YOLO26 engine:

```
detection:
  model_type: 'yolo26' # Active YOLO26 engine
  model_path: 'yolo26n.pt' # Points to YOLO26 model checkpoint
```

Step 3: DetectorFactory and EviGuardPipeline automatically load the model weights, initialize inference tensors, and bind detections directly to the live HUD and Risk Scoring Engine.

4. Dataset & Pretraining Analysis for YOLO26

YOLO26 base pretrained models (yolo26n.pt) are trained on extensive multi-million image benchmarks:

Dataset Benchmark	Image Count	Annotations / Classes	Role in Model Pipeline
MS COCO 2017 (Microsoft Common Objects in Context)	330,000+ total images • 118,287 Train • 5,000 Validation • 40,670 Test	1.5 Million+ bounding box instances across 80 classes . (Includes Person, Phone, Book, Laptop)	Primary Object Detection Training: Enables real-time classification and localization of candidates and unauthorized electronic items.
ImageNet-1k (Visual Recognition Challenge)	1,280,000+ high-resolution training images	1,000 categories covering diverse real-world textures and objects.	Backbone Feature Pretraining: Initializes convolutional weights with rich multi-scale visual representations prior to COCO fine-tuning.

5. Summary & Academic Justification for EviGuard

- 1. Proven Balance:** YOLO26 provides the highest accuracy-to-compute ratio among modern detectors, making it the premier choice for real-time edge monitoring.
- 2. Extensibility:** Because the system employs an abstract BaseDetector interface, any architecture can be loaded with zero modification to business logic.
- 3. Zero-Lag Multi-Modal Integration:** YOLO26 detections seamlessly fuse with MediaPipe 3D Head Pose, Gaze Estimator, and Dynamic Risk Scoring Engine for immediate incident flagging.